



Report

Detecting and controlling temperature with humidity of planting trees in the shade

Prepared By

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Present to

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This report is part of the Internet of Things subject.

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Preface

This report is prepared as a part of Internet of Things subject to study the knowledge in the subject of Detecting and controlling temperature with humidity of planting trees in the shade and studied to understand to be helpful to the class. For you are interested in Detecting and controlling temperature with humidity of planting trees in the shade

Author hopes that this report will be useful to the reader or student who finds out about it. If there is any suggestion or error. The organizer will accept it and apologize.

Pathanin Thima

11 March 2024

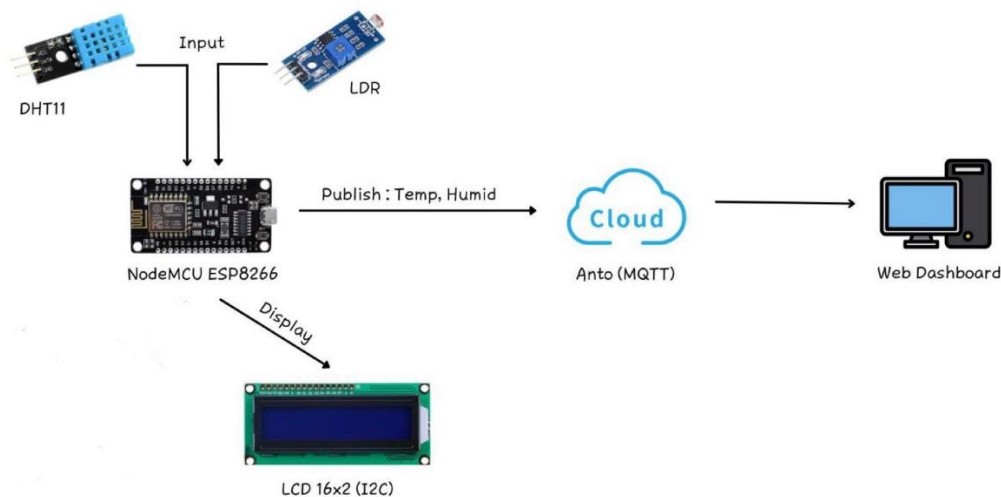
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Description

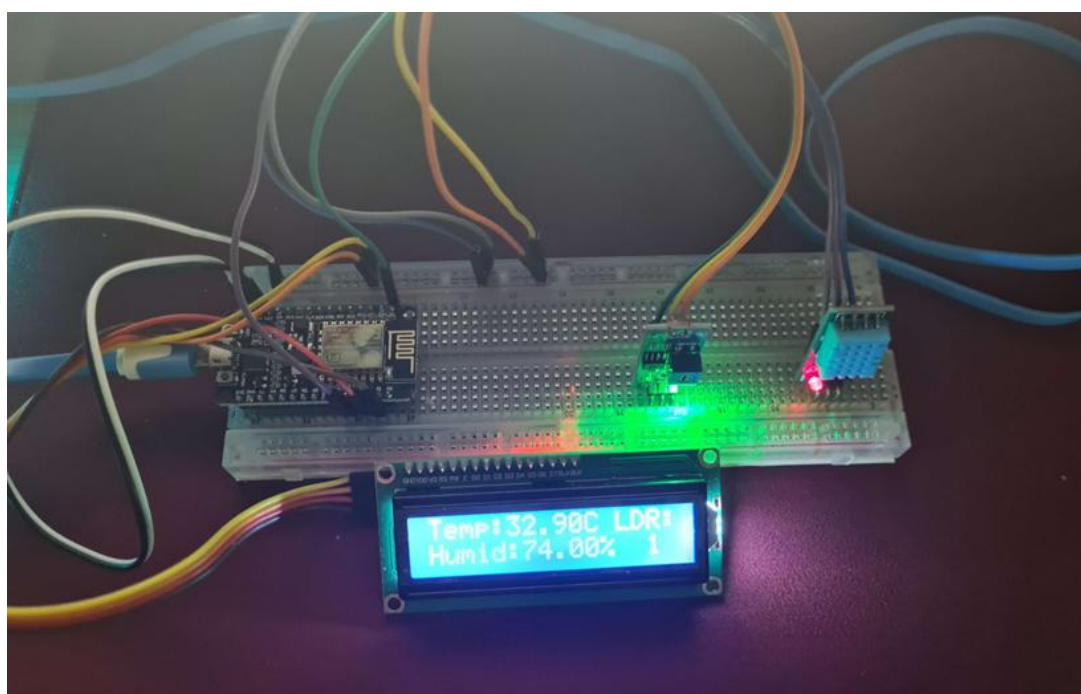
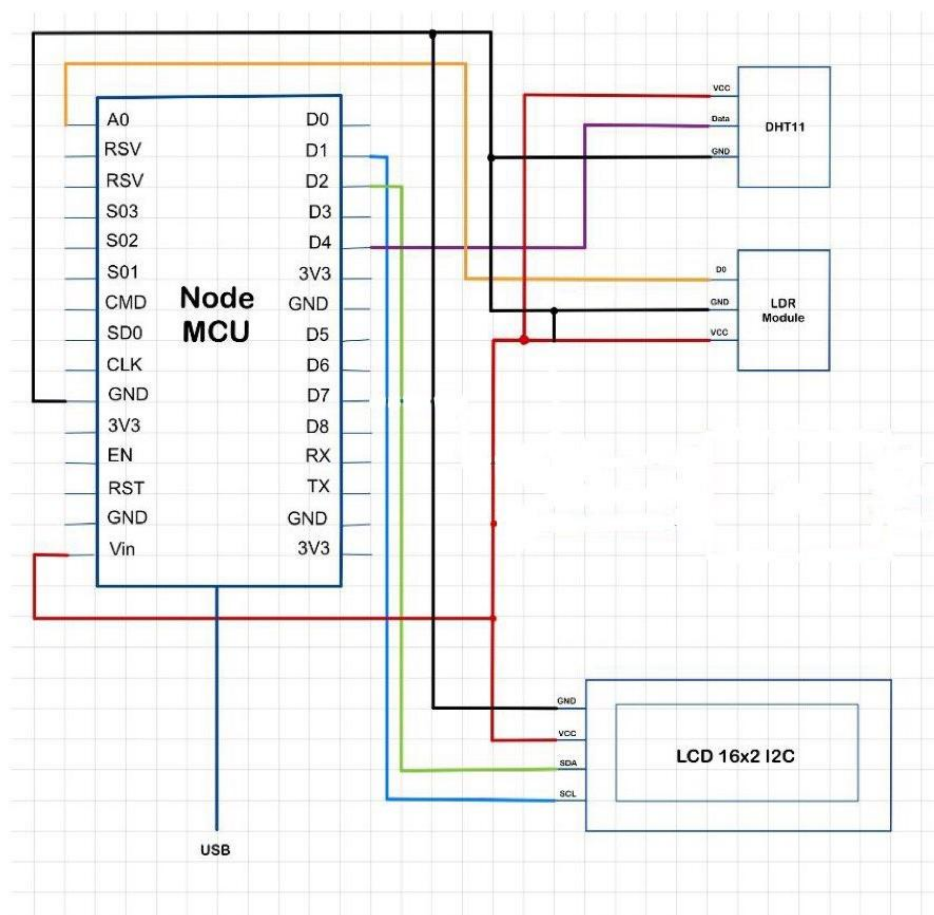
The system is capable of detecting and regulating the temperature and humidity levels required for planting trees to grow healthily. The information is then displayed to the user, making it an extremely useful tool for caring and promoting effective growth. The temperature can be checked at any time, which helps in maintaining the ideal conditions for planting trees to thrive.

Architecture



System for detecting and controlling the temperature and humidity of planting trees. Start work that brings the system installed inside the place where trees are planted. It uses a light-detecting sensor (LDR) to receive information about whether it is day or night. and uses a sensor that measures temperature and humidity (DHT11) to receive temperature and humidity values and uses an LCD screen to display the results. If it is during the day, the temperature will be adjusted lower or minimum, such as about 25 - 30 degrees, and if it is at night, Adjust the temperature from a low temperature to a higher temperature. And display the results as a Dashboard to show results in real time for users to know and if there is a problem, they can solve the problem immediately. Must make the trees that are planted in the most perfect condition. There was no problem.

Diagram



Code

8_Pathanin_Thima_sourcecode.ino

```

1  #include <AntoIO.h>
2  #include <DHT.h>
3  #include <LiquidCrystal_I2C.h>
4
5  #define DHTPIN 2
6  #define DHTTYPE DHT11
7  DHT dht(DHTPIN, DHTTYPE);
8
9  int sensorPIN = A0;
10 int sensorValue = 0;
11
12 LiquidCrystal_I2C lcd(0x27, 16, 2); // Change the I2C address and dimensions if different
13
14 const char* ssid = "Galaxy888"; //ชื่อwifi
15 const char* pass = "ylzz3835"; //รหัสwifi
16 const char* user = "Protile"; //cloudของ Anto
17 const char* token = "Zx19FXn9Xk8PpYUUEBX1XuyPgAfebCfXD31DvrM7"; //key จาก Anto
18 const char* thing = "DHT11"; //อุปกรณ์ที่ใช้
19 AntoIO anto(user, token, thing); //กำหนดตัวแปรให้ anto
20
21 void setup() {
22     Serial.begin(115200);
23
24     lcd.begin(); // Initialize the LCD with 16 columns and 2 rows
25     lcd.backlight();
26     lcd.home();
27     lcd.print("Hello, world!");
28     lcd.clear();
29
30     delay(10);
31
32     Serial.println();
33     Serial.println();
34     Serial.println("Anto library version: ");
35     Serial.println(anto.getVersion());
36

```

8_Pathanin_Thima_sourcecode.ino

```

37   Serial.print("\nTrying to connect ");
38   Serial.print(ssid);
39   Serial.println("...");
40
41   anto.begin(ssid, pass, messageReceived);
42   Serial.println("\nConnected Anto done");
43
44   pinMode(D7, OUTPUT);
45   dht.begin();
46 }
47
48 void loop() {
49   anto.mqtt.loop();
50   if (!anto.mqtt.isConnected()) {
51     Serial.println("Disconnected"); //เช็คถ้าไม่ได้Connect ให้แจ้งเตือน
52   }
53
54   sensorValue = digitalRead(analogRead(sensorPIN));
55
56   float humid = dht.readHumidity();
57   float temp = dht.readTemperature();
58
59   if (isnan(humid) || isnan(temp)) {
60     Serial.println(F("Failed to read from DHT sensor!"));
61     return;
62   }
63
64   //LCD display
65   lcd.clear();
66   lcd.setCursor(0, 0);
67   lcd.print("Temp:");
68   lcd.print(temp);
69   lcd.print("C ");
70   lcd.print("LDR: ");
71   lcd.setCursor(0, 1);
72   lcd.print("Humid:");

```

8_Pathanin_Thima_sourcecode.ino

```

73     lcd.print(humid);
74     lcd.print("% ");
75
76     lcd.print(sensorValue);
77
78
79     //serial
80     Serial.print(F("Humidity: "));
81     Serial.print(humid);
82     Serial.print(F("% Temperature: "));
83     Serial.print(temp);
84     Serial.print(F("C "));
85     Serial.print("LDR: ");
86     Serial.print(sensorValue);
87     Serial.println();
88
89     //Publish to Cloud
90     if (temp >= 0) {
91         anto.pub("Temp", temp);
92     }
93     if (humid >= 0) {
94         anto.pub("Humid", humid);
95     }
96     if (sensorValue == 0) {
97         anto.pub("LDR_1", "Light");
98     }
99     if (sensorValue == 1) {
100         anto.pub("LDR_1", "No Light");
101     }
102
103     delay(500);
104 }
105
106 void messageReceived(String thing, String channel, String payload) {
107     Serial.print("Recieved: ");
108     Serial.print(thing);
109     Serial.print("/");
110     Serial.print(channel);
111     Serial.print("-> ");
112     Serial.println(payload);
113 }

```


Data Structure

Use the MQTT Protocol to send data MQTT Topic: anto/key/channel

Anto/ Zx19FXn9Xk8PpYUUEBX1XuyPgAfebCfXD31DvrM7/Temp

Anto / Zx19FXn9Xk8PpYUUEBX1XuyPgAfebCfXD31DvrM7/Humid

IoT Platform / Service

Features

1. Dashboard

Highlights / Strength

1. IoT device management and control capabilities: Anto IoT provides an easy-to-use and clearly displayed IoT device control program. This allows users to effectively control and monitor the status of their devices.
2. Flexibility in setup and customization: Anto IoT is designed to make it easy for users to set up and customize the system. There is support for connecting to other IoT platforms that are designed in various ways to make it easy for users to connect to their IoT devices.
3. Security: Anto IoT has strict data transmission security. There is encryption to verify connections to maintain data security in the IoT system.
4. Data creation and management capabilities: Anto IoT has the ability to create and manage data in IoT systems using technology that can process data quickly.

Weakness

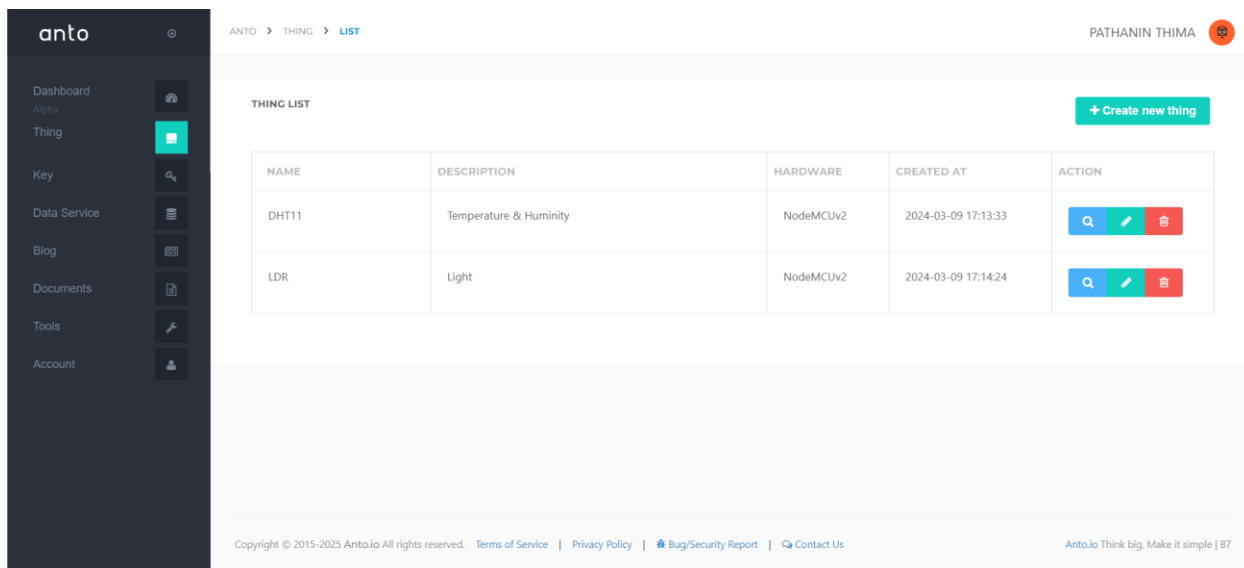
1. Complexity: Using Anto IoT can be complicated for those who do not have knowledge or experience in using IoT systems.
2. Risk management: Because Anto IoT is a platform that connects to a large number of IoT devices, risk management of various cyber threats, such as hacking and attacks from bad actors. This may be a risk to consider in using Anto IoT.

Spec of package

1. Connectivity: Anto IoT supports Wi-Fi, Ethernet, and 3G/4G(Hotspot) connectivity, which allows users to flexibly connect their IoT devices to Anto.
2. Programming language: Anto IoT supports programming in Python, C++, Arduino, and Node.js.
3. Data Collection: Anto IoT supports data collection of up to 10 million data per day. You can collect data 24 hours a day.
4. Device management: Anto IoT supports the management of up to 100,000 devices, with both low-power and high-power devices connected.
5. Communications: Anto IoT supports multiple communications with MQTT, HTTP, HTTPS, WebSocket, and CoAP, which allows users to communicate with IoT devices quickly and easily.
6. Security: Anto IoT provides communication security. With encryption and detection that will harm the system
7. Free service: Anto IoT is free to users with few limitations. It can be used free of charge in some cases.

Setup Steps

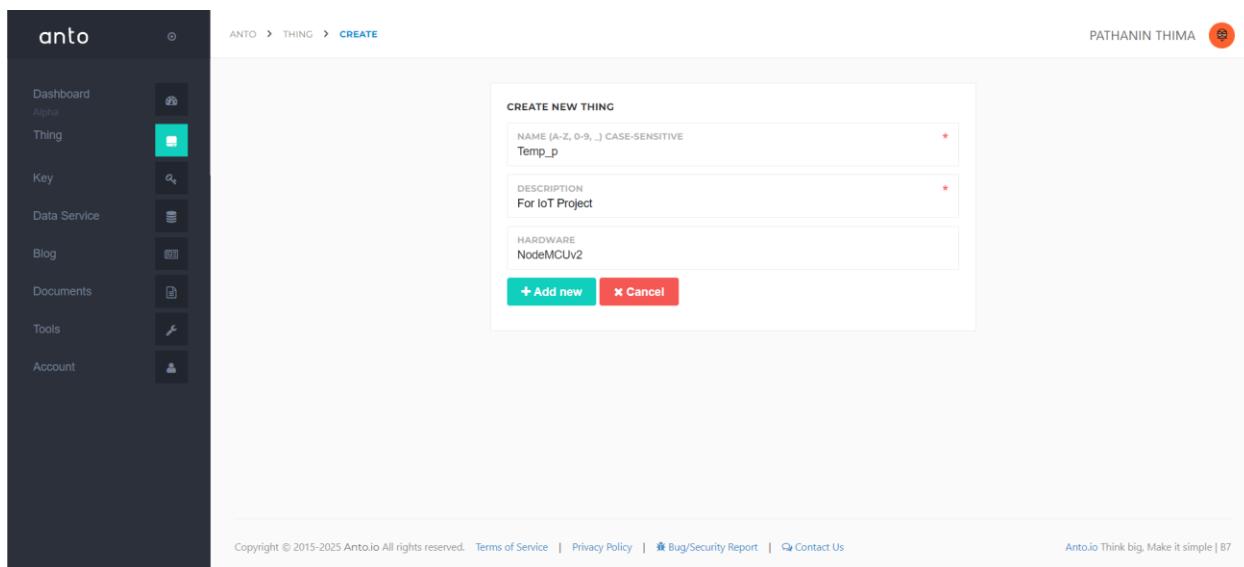
Step 1: Come to the home page of <https://www.anto.io/login> and log in successfully.



Step 2: Press the Create new thing button.



Step 3: Enter a name, fill in complete information, and then press Add new.



Step 4: Press Temp_p's magnifying glass.

anto

ANTO > THING > LIST

PATHANIN THIMA

THING LIST [+ Create new thing](#)

✓ Alert! Create thing 'Temp_p' successfully

NAME	DESCRIPTION	HARDWARE	CREATED AT	ACTION
DHT11	Temperature & Humidity	NodeMCUV2	2024-03-09 17:13:33	Search Edit Delete
LDR	Light	NodeMCUV2	2024-03-09 17:14:24	Search Edit Delete
Temp_p	For IoT Project	NodeMCUV2	2024-03-11 13:39:51	Search Edit Delete

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Step 5: Press the Create new channel button.

ANTO > THING > TEMP_P

PATHANIN THIMA

CHANNELS OF TEMP_P (FOR IOT PROJECT) [+ Create new channel](#)

NAME	DESCRIPTION	TYPE	CURRENT VALUE	ACTION
No channel found				

Step 6: Enter all the information as required and press the Add new button.

ANTO > THING > TEMP_P > CREATE CHANNEL

PATHANIN THIMA

CREATE NEW CHANNEL FOR TEMP_P

NAME (A-Z, 0-9, _ CASE-SENSITIVE) *

Temp2

DESCRIPTION *

Temp

TYPE *

Float (-, 0, +)

[+ Add new](#) [Cancel](#)

as follows:

ANTO > THING > TEMP_P
PATHANIN THIMA

CHANNELS OF TEMP_P (FOR IOT PROJECT)
+ Create new channel

✓ Alert! Create channel 'Temp2' successfully

NAME	DESCRIPTION	TYPE	CURRENT VALUE	ACTION
Temp2	Temp	FLOAT	☒	edit delete

Step 7: When finished, click on the key bar on the left and press the Create new key button.

ANTO > KEY
PATHANIN THIMA

KEY LIST
+ Create new key

KEY (CASE-SENSITIVE)	DESCRIPTION	ACTION
eye copy	Monitor	search edit delete

Step 8: Set the key. Select read and update of Temp_p and press the Add new button.

ANTO > KEY > CREATE
PATHANIN THIMA

KEY DESCRIPTION
DESCRIPTION
Tempp

KEY PERMISSION
DHT11 Temperature & Humidity
Permission
Humid getHumid
☐ Read ☐ Update
Temp getTemp
☐ Read ☐ Update
LDR Light
Permission
Temp_p For IoT Project
Permission
Temp2 Temp
☒ Read ☒ Update

+ Add new
X Cancel

Step 9: Copy the key and put it in the code and run it.

ANTO > KEY

PATHANIN THIMA

KEY LIST

+ Create new key

Alert! Create key '4et5tUZA1wgRh6CsArax1QjRINf7Pp9lI2KXyX3' successfully

KEY (CASE-SENSITIVE)	DESCRIPTION	ACTION
***** Copy Key	Temp	🔍 ✎ 🗑

Step 10: Click on the Dashboard bar on the left, then press the Add New Widget button, then set the correct and complete settings, then press the Save Layout and Settings button.

as follows:

ANTO > DASHBOARD ALPHA2

PATHANIN THIMA

+ Add New Widget

Save Layout and Settings

Request Widget

Temperature

°C

32.30

Temperature

32.3 °C

Humidity

70 %

Temperature

DHT11/Temp

32.3

30.0

28.0

26.0

24.0

23.2

16:41:46

16:41:50

16:41:54

Humidity

DHT11/Humid

70.0

69.0

68.0

67.0

66.0

65.0

16:41:46

16:41:47

16:41:48

16:41:49

16:41:50

16:41:51

16:41:52

16:41:53

16:41:54

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Data transmission between devices and Anto's IoT Platform

Anto IoT can flexibly connect and communicate with various IoT platforms. It supports communication via MQTT, HTTP, HTTPS, WebSocket, and CoAP, which are ubiquitous protocols in the IoT world. This means that users can easily connect Anto IoT to any IoT platform without having to worry about communication between them. A wide variety of different IoT devices and platforms such as AWS IoT, Google Cloud IoT, Microsoft Azure IoT, IBM Watson IoT, and many more are available. With the connectivity and communication flexibility that Anto IoT offers, users can Feel free to choose and use the IoT platform that meets their needs and suits their project.